Interview Memo - Expert Interviews on Residential Window Repair and Component Replacement

Sources included:
INT-01 - Window supplier
INT-02 - Architect

Purpose:
This memo synthesises the two exploratory expert interviews and connects the practical findings to the literature coding, the VELFAC 200 ENERGY window-system case, and the later risk pathways.

Main pattern:
The main pattern across both interviews is that window repairability is not only a property of the window product itself. It depends on whether short-life components can be accessed, whether compatible parts remain available, how the window is installed into the wall, whether dimensions are standardised, and whether architects, suppliers, clients and regulations support repair-oriented decisions. Both interviews show that component-level repair and replacement can be technically possible, but the practical conditions around the product often determine whether repair actually happens.

Most common codes:
Component accessibility / Component replacement / Repair / Design intervention / Maintenance planning / Spare parts / Implementation gap / Premature replacement / Direct reuse / Lifecycle verification / Standardisation / Lock-in

Where the experts agreed:
Both experts identified opening mechanisms, hardware, handles or related functional parts as components that tend to wear or fail before the full window frame. Both also confirmed that certain components, such as hardware, handles, fittings and glazing units, can often be repaired or replaced without replacing the complete window. This supports the project's focus on retaining long-life components while maintaining or replacing shorter-life parts.

Both experts also pointed to access as a major condition for repairability. The supplier focused on physical access to hinges, mechanisms and other serviceable components, especially where window reveals or finishes make repair difficult. The architect expanded this point by linking repairability to the wall/window interface and installation strategy, including whether future removal, access or replacement has been considered during design and specification.

Both experts also described direct reuse as difficult. The supplier pointed to project-specific window dimensions and ageing components, while the architect highlighted the lack of standardised dimensions in the Bulgarian context. Together, these findings suggest that theoretical reuse potential does not automatically become practical reuse unless dimensional compatibility, certification, service infrastructure and future demand are addressed.

Where the experts differed:

The supplier gave more practical information about common failures, repairable parts, spare-part availability and client preference for newer or more modern window systems. His answers were strongest for understanding repair barriers at the component and service level.

The architect gave a broader design and specification perspective. He emphasised installation method, dimensional standardisation, client briefs, supplier information flow and the absence of stronger regulatory or standardisation requirements. His answers were strongest for understanding how architectural decisions and project conditions affect repairability before the window is even installed.

The two interviews also differed slightly on supplier information. The supplier did not provide detailed environmental or verified service-life information, while the architect stated that suppliers can usually provide more information if it is specifically requested. This suggests that the issue may not be only the absence of information, but also whether the right information is requested, transferred and used at the correct stage of design.

Contradictions or surprises:
Neither expert provided a direct example of a sustainable-looking or energy-efficient window decision that later created an unexpected problem. This means the interviews provide stronger evidence for implementation gaps, repairability barriers and lifecycle uncertainty than for direct rebound effects. This should be treated carefully: it does not prove that rebound or unintended consequences do not occur, only that these two interviews did not provide direct examples.

A useful contrast also appeared around modularity and high-performance systems. The supplier did not identify replaceable parts as a direct cause of technical complexity or more difficult maintenance. The architect, however, noted that higher-performance or more advanced systems may involve higher cost, more material and more complex mechanisms, even if they may also provide longer service life and better energy performance. This supports the need for lifecycle verification rather than assuming that either simpler or more advanced systems are automatically better.

Most important practical insight:
The most important practical insight is that repairability must be designed and specified as a system condition, not assumed from the existence of replaceable parts alone. A window can contain repairable components, but repair may still fail in practice if access is blocked, parts are unavailable, the window is embedded into the wall assembly, dimensions are too project-specific, or the client chooses replacement for perceived improvement rather than technical necessity.

Link to VELFAC 200 ENERGY case:
The interviews strengthen the VELFAC 200 ENERGY case by showing why product documentation alone is not enough to judge product-life extension potential. The VELFAC case includes a wood-aluminium frame, triple glazing, selected maintenance and replacement guidance, and an EPD, but the interviews show that practical repairability also depends on access to components, spare-part availability, installation method, dimensional compatibility and information flow between supplier, architect and client.

The supplier interview is especially relevant to the VELFAC case because it highlights the importance of hardware, handles, glazing units, seals and access to serviceable parts. The architect interview is especially relevant because it shows that even a technically high-quality window system may become difficult to repair or replace if the installation strategy and surrounding construction do not allow future access.

Link to risk pathways:
The interview findings support several later risk pathways:

1. Technically repairable window → poor access to hinges, mechanisms or serviceable parts → repair becomes impractical → implementation gap → design for component access and maintenance clearance.

2. Replaceable components → compatible parts unavailable after long use period → complete window replacement → implementation gap or premature replacement → supplier commitment and long-term spare-part planning.

3. High-performance or more complex window system → higher initial cost, more material and more complex mechanisms → uncertain lifecycle benefit → lifecycle trade-off → lifecycle verification before specification.

4. Custom-sized window → difficult direct reuse in another project → reuse potential not realised → implementation gap or burden shifting → dimensional standardisation and local reuse infrastructure.

5. Client preference for newer or more modern windows → full replacement despite repair possibility → premature replacement or user-driven obsolescence → maintenance hierarchy and repair-first decision guidance.

6. Window embedded into wall assembly → difficult future removal or component access → technological lock-in → installation strategy that allows future repair, adjustment or replacement.

Limitation:
The interviews represent two exploratory expert perspectives and are not statistically representative or validated. They are used to support practical interpretation of the literature findings, not to prove prevalence or causal magnitude. The supplier interview was anonymised, while the architect interview was used with permission including name and organisation. Both interviews were short and focused on practical knowledge. Neither interview provides quantified lifecycle data, verified environmental impacts or market-level evidence of rebound effects.

Short synthesis paragraph:
The two expert interviews confirm that the environmental value of repair and component replacement depends on practical enabling conditions. Both experts agreed that shorter-life components such as opening mechanisms, handles, fittings and glazing units can often be repaired or replaced without replacing the whole window. However, repairability depends strongly on access, installation method, compatible spare parts, dimensional standardisation, supplier information, client expectations and wider regulatory or specification requirements. The interviews therefore support the project's distinction between technical repairability and practical implementation. They also show why a preliminary screening framework should ask not only whether a component can theoretically be replaced, but whether the design, installation, supply chain and decision-making context make repair likely to happen in practice.